PART ONE

MATH 152 **EXAM II** NAME _____

Mrs. Bonny Tighe 50 points

7.4 -11.3 TA_____ Wed. 10/25/2017

1. (5 points) Complete the following tests

1GEOMETRIC SERIES $\sum_{n=1}^{\infty} a_n (r)^{n-1}$ _____ if $|r| \geq 1$

TEST FOR DIVERGENCE **If** $\lim_{n\to\infty} a_n \neq$ _____ , then $\sum_{n=1}^{\infty} a_n$ is divergent

THE INTEGRAL TEST Suppose f is a continuous, positive, decreasing function on $[1, \infty)$ and let $a_n = f(n)$ then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if _____is convergent.

P-SERIES $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is _____ if $p \leq 1$

REMAINDER ESTIMATE FOR THE INTEGRAL TEST
Suppose $a_k = f(k)$ where f is a continuous, positive, decreasing function for $x \geq n$, and $\sum a_n$ is convergent . If $R_n = S - S_n$, then $R_{n+1} \leq$ _____

2. (10 points) a) Find a formula, a_n, for the general term of the sequence. Is the sequence convergent or divergent? Explain your answer?

$$\left\{\frac{1\cdot 5}{1}, \frac{1\cdot 5\cdot 9}{2}, \frac{1\cdot 5\cdot 9\cdot 13}{6}, \frac{1\cdot 5\cdot 9\cdot 13\cdot 17}{124} \ldots\right\}$$

b) Express $0.0\overline{24}$ as a ratio of integers using infinite series.

3. (10 points) Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $\sum_{k=2}^{\infty} \frac{1}{k^2 + 2k - 3}$

b) $\sum_{k=0}^{\infty} \frac{2^{k+1}}{e^{k+2}}$

4. (10 points)

a) Use the Integral test to find if the following series is convergent or divergent.

$$\sum_{n=0}^{\infty} \frac{e^n}{e^{2n} + 1}$$

5. (10 points) $\int \frac{x^3}{x^2 + x - 6}\, dx$

6. (10 points) $\int \frac{x^2}{\sqrt{25 - x^2}}\, dx$

PART TWO

MATH 152 **EXAM IIA** NAME _____

Mrs. Bonny Tighe 50 points

7.4-11.3 TA_____ Wed. 10/25/2017

1. (10 points) Are the following improper integrals convergent or divergent? Describe what convergence means.

a) $\int_0^3 \frac{4x}{\sqrt{9-x^2}}\,dx$

b) $\int_1^\infty x^2 e^{-x^3}\,dx,$

2. (10 points) Find the length of the curve $x = \ln(y^2 - 1)$ $2 \le y \le 3$

3. (15 points) a) The given curve is rotated about the x-axis. Find the area of the resulting surface. $S = 2\pi \int_a^b radius \;\; arclength$ $\qquad y^{2/3} + x^{2/3} = 9, \;\; 1 \le x \le 8$

4. (10 points) a) How large should we take n in order to guarantee that the Midpoint approximation for $\int_2^4 (\ln x)dx$ is accurate to within 0.01? (see next page)

b) Estimate the error in using the partial sum S_{10} of the series $\sum_{k=1}^{\infty} \frac{1}{k^4}$

5. (10 points) Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for the Midpoint rule. Suppose $|f''(x)| \le K \quad for \quad a \le x \le b$, then

$$|E_M| \le \frac{K(b-a)^3}{24n^2} \qquad \int_1^2 \frac{1}{e^x}dx, \quad n = 4$$

PART ONE

MATH 152 **EXAM II A** NAME _____

Mrs. Bonny Tighe 50 points

7.4 -11.3 TA_____ Wed. 10/25/2017

1. (5 points) Complete the following tests

1GEOMETRIC SERIES $\sum_{n=1}^{\infty} a_n (r)^{n-1}$ _____ if $|r| < 1$

TEST FOR DIVERGENCE **If** $\lim_{n\to\infty} a_n \neq$_____ , then $\sum_{n=1}^{\infty} a_n$ is divergent

THE INTEGRAL TEST Suppose f is a continuous, positive, decreasing function on $[1, \infty)$ and let $a_n = f(n)$ then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if _____is convergent.

P-SERIES $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is _____ if $p > 1$

REMAINDER ESTIMATE FOR THE INTEGRAL TEST
Suppose $a_k = f(k)$ where f is a continuous, positive, decreasing function for $x \geq n$, and $\sum a_n$ is convergent . If $R_n = S - S_n$, then $R_{n+1} \leq$_____

2. (10 points) a) Find a formula, a_n, for the general term of the sequence. Is the sequence convergent or divergent? Explain your answer?

$$\left\{\frac{1\cdot 3}{2}, \frac{1\cdot 3\cdot 5}{6}, \frac{1\cdot 3\cdot 5\cdot 7}{24}, \frac{1\cdot 3\cdot 5\cdot 7\cdot 9}{120}\ldots\right\}$$

b) Express $0.0\overline{47}$ as a ratio of integers using infinite series.

3. (10 points) Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $\sum_{k=2}^{\infty} \frac{1}{k^2 + k - 2}$

b) $\sum_{k=0}^{\infty} \frac{2^{k+2}}{e^{k+1}}$

4. (10 points)

a) Use the Integral test to find if the following series is convergent or divergent.

$$\sum_{n=0}^{\infty} \frac{e^n}{e^{2n} + 1}$$

5. (10 points) $\int \frac{x^3}{x^2 - 3x - 4}\, dx$

6. (10 points) $\int \frac{x^2}{\sqrt{9 - x^2}}\, dx$

PART TWO

MATH 152 **EXAM II** NAME _____

Mrs. Bonny Tighe 50 points

7.4-11.3 TA_____ Wed. 10/25/2017

1. (10 points) Are the following improper integrals convergent or divergent? Describe what convergence means.

a) $\int_0^2 \frac{3x}{\sqrt{4-x^2}}\,dx$

b) $\int_1^\infty x^3 e^{-x^4}\,dx,$

2. (10 points) Find the length of the curve $y = \ln(x^2 - 1)$ $2 \le x \le 3$

3. (15 points) a) The given curve is rotated about the x-axis. Find the area of the resulting surface. $S = 2\pi \int_a^b radius \;\; arclength$ $\qquad y^{2/3} + x^{2/3} = 4, \;\; 1 \le x \le 2$

4. (10 points) a) How large should we take n in order to guarantee that the Midpoint approximation for $\int_1^2 (\ln x)dx$ is accurate to within 0.001? (see next page)

b) Estimate the error in using the partial sum S_{20} of the series $\sum_{k=1}^{\infty} \frac{1}{k^3}$

5. (10 points) Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for the Midpoint rule. Suppose $|f''(x)| \le K \quad for \quad a \le x \le b$, then

$|E_M| \le \frac{K(b-a)^3}{24n^2}$ $\qquad \int_1^2 \sqrt{x}\, dx, \quad n = 4$